

Lab 6

Introduction to Databases



Join tables and Retrieving data from more than one table

1. Create a two Tables:

Create a table Supplier with the following columns:

- supplier_id → NUMBER (Primary Key)
- supplier_name → VARCHAR2(50) Not Null

Create a table Product with the following columns:

- product_id → NUMBER
- product_name → VARCHAR2(50) Not Null
- supplier_id → NUMBER
- Price → NUMBER

2. Insert Data into a Table:

Insert the following Supplier into the table:

(100, 'IBM')
(200, 'HP')
(300, 'Microsoft')
(400, 'Apple')

Insert the following Product into the table:

(1, 'IPAD2', 400, 2400)
(2, 'IPHONE 4s', 400, 2500)
(3, 'MS Office 2012', 300, 1600)
(4, 'Color Printer', 100, 1500)

3. Retrieving the supplier_name , product_name, Price using **Join -- On**
4. Retrieving the supplier_name , product_name, Price using **Where**
5. Retrieving the supplier_name , product_name, Price **Where price more than 1000**
6. Retrieving the supplier_name , Price **Where product name is Apple**

SELECT **AggregateFunctionname (column name)** FROM table WHERE conditions;

7. Find the total number of products in the Product table.
8. Find the average price of all products.
9. Find the most expensive and the cheapest product.
10. Find the total sum of all product prices.
11. Find the total price of products whose price is above 2000.
12. Display supplier names in uppercase
13. Display supplier names in lowercase

Combine multiple aggregation functions with GROUP BY and HAVING.

Before join tables (No Join):

- 14. Count products per supplier id
- 15. Display the Suppliers id with more than or equal 2 products
- 16. Count products for each supplier where price > 500, and show only suppliers with at least 2 products.

After join tables (join):

- 16. Find the average product price for each supplier.
- 17. Find the highest priced product for each supplier.
- 18. Find how many products each supplier provides.
- 19. List suppliers who supply more than one product.

Home Work

A university system where: A **Student** can enroll in many **Courses** A **Course** can have many **Students**

Given Tables

Student (student_id, student_name)

Course (course_id, course_name, credit_hours)

Enrollment (student_id, course_id, grade)

Student

student_id	student_name
1	Ali
2	Sara
3	Omar
4	Mona

Course

course_id	course_name	credit_hours
101	Database	3
102	Java	4
103	Networks	3
104	AI	2

Enrollment

student_id	course_id	grade
1	101	85
1	102	78
2	101	90
2	103	88
3	102	70
3	104	95
4	101	60
4	103	75
4	104	80

1. Write a query to display: student_name, course_name and grade
2. Write a query to display all students with the courses they are enrolled in.
3. Write a query to find the **number of courses each student is enrolled in.**
4. Write a query to find the **average grade for each student.**
5. Write a query to find the **total number of students in each course.**
6. Write a query to find the **maximum grade in each course.**
7. Write a query to display students who are enrolled in **more than 2 courses.**
8. Write a query to display students whose **average grade is greater than 80.**
9. Write a query to display courses that have **more than 2 students enrolled.**
10. Write a query to find the **average grade for each course**, and display only courses with average grade greater than 75.